A.I. Matrix: Authentic v. Synthetic Image Generation

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Abstract

The rapid advancement of artificial intelligence (AI) has made image generation easily accessible and easy to use, but the effectiveness of different types of training data remains a critical question. This project explores whether synthetic (artificially generated) data can rival and/or outperform authentic (real-world) data in training AI image generation models. The research question focused on identifying any statistical difference in performance between models trained on authentic versus synthetic datasets.

Two datasets were created: one synthetic and one authentic, each containing 200 images. Both AI models were prompted with the phrase, “Image of a girl with blonde hair,” and the resulting outputs were evaluated using a standardized Likert scale ranging from 1 (horrible) to 5 (excellent). The scores were based on the accuracy and performance of the output images in comparison to the prompt. Results revealed that the synthetic dataset outperformed the authentic dataset, with an average score of 3.12 compared to 2.72 for the authentic model. The standard deviation of ±1.05 for both datasets indicated similar levels of variability in quality, confirming the reliability of the comparison.

The findings suggest a potential advantage in using synthetic data to train A.I. models, particularly in fields where authentic data is difficult, expensive, or unethical to collect. This study concludes that synthetic training data is not only a viable alternative but, in some cases, a superior option to authentic data for A.I. image generation. These results have significant implications for industries relying on AI, where access to high-quality, ethical, and scalable training data is essential.

Keywords: Likert scale, dataset, synthetic, performance, accuracy

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Literature Review

Introduction

Artificial Intelligence (AI) generative image models are trained off of datasets consisting of images scraped off of the internet. Artificial Intelligence models utilize artificial neural networks, a digital model of the human brain. AI models are trained using a combination of techniques, the most common of which are 1.) Supervised learning, 2.) Unsupervised learning, and 3.) Reinforcement learning. Supervised learning consists of datasets matched with detailed text descriptions, unsupervised learning allows the model to reconstruct images through unlabeled datasets, and reinforcement learning fine-tunes itself according to human feedback. Training neural networks on datasets with tens of millions of works "enables AI image generators to transform language into realistic visual media"(NightCafe, 2023). Datasets used for training can violate copyright claims because AI has the power to recreate the works using the original style. Similarly, AI users struggle to gain copyright for their works in the U.S. because under U.S. copyright law, copyright protection does not extend to non-human creators so courts have "a limited ability to impose an industry-wide solution" (Hodson, 2023) regarding copyrighted works protection from AI. Large companies have been dominating the AI landscape "with Microsoft-backed OpenAI and other startups raising large sums of money" and how the "combined corporate value of generative AI companies jumped sixfold in two years to $48 billion"(Takeda 2023). This review will focus on the legal issues with AI, and how AI impacts artists.

How do AI Image Generators' Training Interfere with Copyright?

The datasets used to train AI often contain copyrighted works, a result of the data collection process. U.S. Copyright law gives authors exclusive rights to reproduce, adapt, display, and

perform the original work. AI developers use the Fair Use doctrine to defend against copyright claims. The Fair Use Doctrine promotes freedom of expression by permitting the unlicensed use of copyrighted works depending on four factors: potential commercial purpose, nature of the copyrighted work, amount of original work used, and the prospective value on the market. Jenny Quang focuses on the legal issues regarding AI with the goal of providing reasoning for a safe harbor for training data, and Goel et al. focus on the technical side to determine the accuracy of AI.

Copyright Infringement and AI Image Generators

Copyrighted work used in AI image generator datasets for training has sparked court cases based on copyright infringement. Jenny Quang analyzed the debates and outcomes of court cases involving AI developers and owners of copyrighted works, concluding Congress should adopt " a safe harbor for data mining"(Quang, 2021, p. 1407). Quang determined it to be inconvenient for AI developers to "determine what is copyrighted, and request permission from each creator" when datasets contain tens of millions of works(Quang, 2021, p. 1429). Quang theorized the best defense against copyright claims in AI datasets is the Fair Use Doctrine, which provides a degree of flexibility when it comes to the U.S. Copyright law. Copyright has become a potential barrier when machine learning models are using expressive data, so the Fair Use Doctrine can act as a replacement. Supreme Court case Baker v. Selden ruled data mining for training AI would fundamentally not be copyright infringement because it "requires not just copying of a work's material form but also the unauthorized use of the work for its expressive purpose"(Quang, 2021, p. 1408). If the usage of the work is merely technical and non-communicative, it cannot be considered copyright infringement.

An artificial neural network is a collection of simple interconnected algorithms processing information in response to external input, for example, giving a prompt for an AI image generator and the neural network processes the prompt and gives a response. Of the three commonly used ways of training an AI, Goel et al. analyzed the role of "artificial neural networks and machine learning in utilizing spatial information"(2023, p. 275) and the applications of Artificial Intelligence, examples including medical diagnosis, facial recognition, weather forecasting, and more. From the data they were able to determine the accuracy and legitimacy of the data used by the AI. Goel et al. analyzed machine learning, artificial neural networks, and the deep learning methods used on the AI. Goel et al. concluded the world of AI, due to "extensive use and progress of digital data, the enlargement of computing power"(2023, p. 277) will continue to expand and the applications of it will grow based on their research of current applications and the accuracy of these applications in the real-world environment. Goel et al. also determined unsupervised learning is the best for AI image generators because it is best if it "does not have labeled data since the data is not implicitly classified into distinct groups" and the "prediction model learns the data by detecting implicit patterns"(2023, p. 277). After all, the datasets are based on images. Goel et al. determined "facial recognition, and weather forecasting, to name a few, have all seen a surge in interest in the last few years"(2023, p. 277) due to artificial neural networks. Regarding facial recognition, it is one " of the greatest effective and relevant usages of biometric and image processing"(Goel et al., 2023, p. 278). Facial Recognition Systems match the human face and compare it to computer photos, used in offices for discriminating records. A convolutional neural network (CNN) is a type of artificial neural network used primarily for image recognition and processing, and "has been used extensively for image processing and facial identification at different orientations"(Goel et al., 2023, p. 278).

Synthesis

Both Jenny Quang and Goel et al. analyzed the impact of AI, but Quang focused on legal challenges and copyright while Goel et al. focused on the technical and practical applications of AI. Copyright and data integrity can affect the development of AI, with Quang arguing that copyright can be a potential barrier for development. The Fair Use Doctrine is a safeguard against copyright and mitigate restrictions. Similarly, Goel et al. emphasize the importance of accurate data, with AI development being dependent upon the data. Both from the legal side and technical side, Quang and Goel et al. emphasize the importance of legal clarity and data quality for AI development. Quang and Goel et al. diverge in opinion regarding the future of AI, with Quang being concerned with how without legislative reforms, U.S. AI development will fall behind, with copyright being to blame. Goel et al. take a more optimistic approach with how AI applications are becoming increasingly accurate with examples being facial recognition and weather forecasting. Despite these differences, both Quang and Goel et al. agree on the power of AI, with Quang's proposal for a "safe harbor for data mining" aligns with Goel et al. analysis of how CNNs and facial recognition systems are effective tools for processing datasets. What stands out about both sources is the tension between AI innovation and structural systems, legal and technical. While Quang's analysis of court cases advises legislative action to boost AI development, Goel et al. highlights the promise of AI built on trustworthy systems. Together they show the challenge of fostering innovation while maintaining ethical and legal integrity, which will shape the future of AI.

How does AI Image Generation affect Artists?

With the recent use of advanced AI models capable of mimicking human creations, it has become a valid possibility that AI creations might soon replace products of human creativity. It

has been predicted that AI could not only produce art independent of human creativity but also assist human efforts, leading to an increase in artistic quality (Zhou & Lee, 2024, p. 52). There is no way to indicate if this event will indeed forecome, but there are arguments on why it also may not. One possible reason artistic replacement by AI may be unlikely is how, independent of the physical properties of art, we place great value on the imbuement of the human experience in art, and people view this imbuement as a necessary part of artistic creation (Bellaiche et al., 2023, p. 42). Thus, the idea that AI productions are not "real" pieces of art, a definition of validity based on perception, has become rather popular among mainstream media.

Creative Disputes and Perception

"Art is widely considered to be a uniquely human phenomenon" (Bellaiche et al., 2023, p. 42). Art is known to encapsulate emotions, express individualistic experiences, and serve as commentary, all of which are elements commonly thought to be human-specific. Overall, it is thought people place great value on the imbuement of the human experience in art. To explore this concept, Bellaiche et al. manipulated the purported creator of pieces of art by randomly assigning a "Human-created" or "AI-created" label to paintings all actually created by AI and then assessed participants' judgments of the artworks. Despite all 30 of the paintings being created by AI, those labeled as being created by a human consistently scored higher among the study participants. Bellaiche et al. determined this to mean people tend to be negatively biased against AI-created artworks relative to purportedly human-created artwork and suggest knowledge of human engagement in the artistic process contributes positively to appraisals of art. For instance, Belliache et al.'s data could suggest not only is AI art liked less, but it is also viewed as less worthy and less profound, which may have interesting implications for the ways in which people will consume AI art in the future.

Though AI-generated art is generally seen in a negative light, it does significantly reduce the time artistic creation can take while also possibly inspiring ideas in the user. Thus, the question of how "the adoption of generative AI [can] affect humans' creative production" becomes increasingly more relevant (Zhou & Lee, 2024, p. 52). Zhou and Lee conducted multiple separate analyses of this question in search of a common link. Zhou and Lee's research models how, over time, text-to-image AI significantly enhances human creative productivity by 25% and increases the likelihood of favorable reception by nearly 50%. Importantly, the data suggests AI-assisted artists who can successfully explore more novel ideas, regardless of prior originality, produce artworks peers evaluate more favorably. Zhou and Lee results suggest ideation and filtering are necessary skills in the text-to-image process, thus giving rise to "generative synesthesia"—the harmonious blending of human exploration and AI exploitation to discover new creative workflows.

Impact and Importance

Multimodal generative AI systems with text, images, videos, voice, and music as inputs and/or outputs have quickly proliferated into the mainstream. Notably, generative AI products have entered the market, tools like Midjourney, Stable Diffusion, and DALL-E, having tens of millions of daily users, making AI image generation an estimated $48B industry (Kentaro, Akira, and Yukiko, 2023). However, professional artists have spoken up about the harms they have experienced due to the proliferation of large-scale image generators, chief among them being economic loss from automating the artistic process. (Jiang et al., 2023, pp. 363 - 374).

Synthesis

The perception and impact of AI-generated art have become underlying themes in the discourse surrounding its usage. A key area of these themes is the changing definitions of creativity and

authorship, as most are conflicted on the validity of AI art. Some view it to be an extension of human ingenuity, where the machine just serves to assist artistic creation (Zhou & Lee, 2024, p. 52). AI art has the possibility to significantly reduce the time artistic creation can take while also possibly inspiring ideas in the user, given they are seeking inspiration and not work. Others skeptically question its authenticity, debating over the significance of art created without human intent. People tend to be negatively biased against AI-created artworks relative to purportedly human-created artwork, suggesting knowledge of human engagement in the artistic process contributes positively to appraisals of art (Belliache et al., 2023, p. 42). Overall, the perception of AI-generated art is varied, greatly so, and a middle ground is nearly nonexistent. Though, if summarized in layman's terms, it would seem people view AI art as a “shortcut”, where it undermines the beauty of the human experience, while also disregarding its potential use as a tool to assist the process of artistic creation, possibly enhancing the original artwork or ideal. Despite this, AI image generation continues to evolve as an industry, affecting more people by the hour (Jian et al., 2023, pp. 363 - 374).

Discussion

Collecting data on the different aspects of AI art, whether legal or artistic, is difficult, as most studies on this topic are observational instead of experimental. This is why there is plenty of information on what AI art is, how it functions, and how it is created, but few sources on its ramifications. The lack of clear legal ownership creates challenges for regulating the use and distribution of AI art. Potential copyright infringement in datasets has become a prolific issue, asserting artists are not getting fair compensation for their use in training. Additionally, AI-generated art has disrupted traditional art markets, strongly disturbing the economic security of artists by shaking the value of human creativity. It is due to these aspects people view AI art

with a negative lens, creating a negative stigma over the use of a tool. In order to account for the rapid expansion of this growing industry, legal frameworks need to be adapted for the use of AI generation and the Art community must navigate through rapid technological advancement and change. AI-generated art may disrupt traditional notions of authorship and human creativity while posing a risk to the livelihood of artists, but it can also introduce novel forms of artistic expression and assist artists in the pursuit of new media. AI art represents both a challenge and an opportunity.

One-Pager

Research Question

How does whether or not the image is synthetic or authentic impact an AI image generator's accuracy and performance for image generation?

Aim

Determine the differences in AI image generator frameworks trained on synthetic images versus authentic images.

Objectives

Identify biases and inaccuracies in AI Image generators trained off of synthetic images.

Analyze the comparative effectiveness of synthetic image data and real-world data in training AI Image models.

Train an AI model framework to compare synthetic and authentic image generator images.

Background and Significance

The growing use of synthetic images for training purposes in AI image generator models offers advantages and challenges. Synthetic datasets provide a scalable alternative to real-world data collection but often lack the accuracy needed for user prompts. Studies, such as those from Nature Machine Intelligence and MIT CSAIL highlight how models trained solely on synthetic data face issues related to performance for real-world applications. Also, platforms like Omnilert and the United Nations University caution against using synthetic data because it can amplify biases and misrepresent certain populations, which lead to ethical concerns.

Methods

Our project is an explanatory case study on comparing the accuracy and performance of synthetic and authentic images. By seeking out contacts to refer to and learn from, we will gather the information we need to learn how to train our own AI model framework and analyze the comparative effectiveness of synthetic image data and real-world data in training AI image models. To analyze the effectiveness of the AI models, we will compare the output's accuracy to the prompt, and identify any potential biases in the outputs.

Mentors and Contributors

Zach Chester

Raw Data

"Fake" Dataset comprised only of Synthetic Images (the first two hundred images)
https://huggingface.co/datasets/TLeonidas/this-person-does-not-exist/discussions
"Real" Dataset comprised only of Authentic Images (the first two hundred images)
https://huggingface.co/datasets/HK83/real_people_3000

Data Analysis Assignment

Raw Data - Fake vs Real Faces - Taking 200 images for each dataset

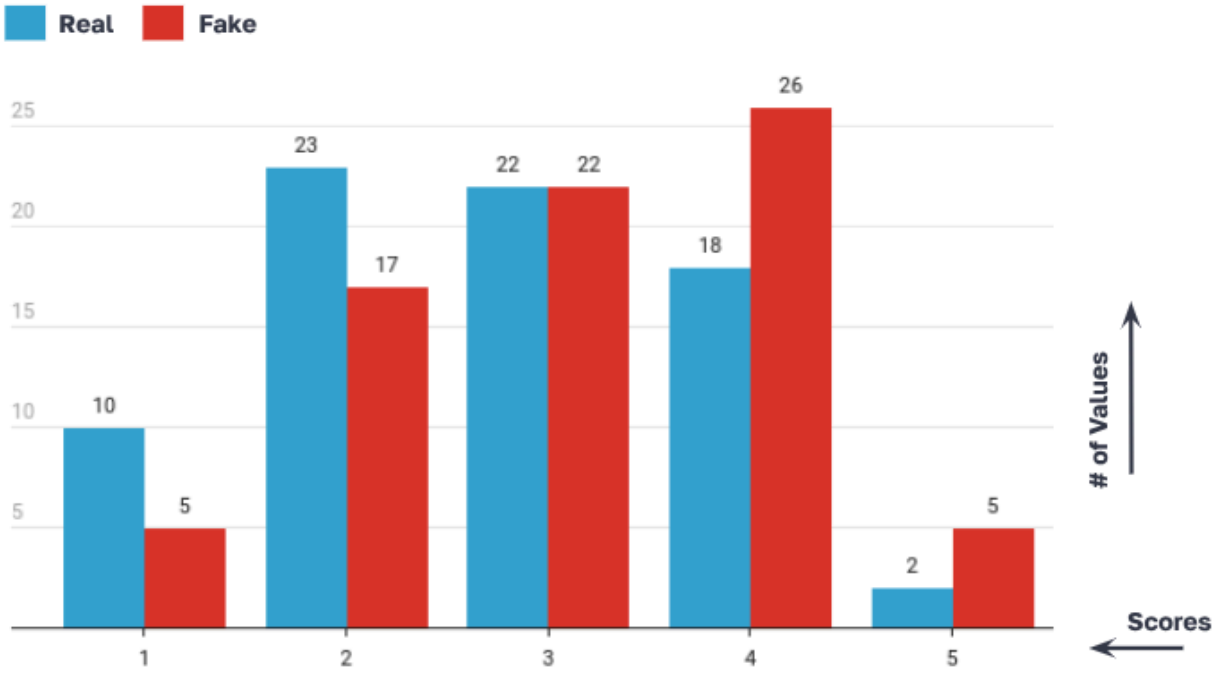
Real - 200 images

Fake - 200 images

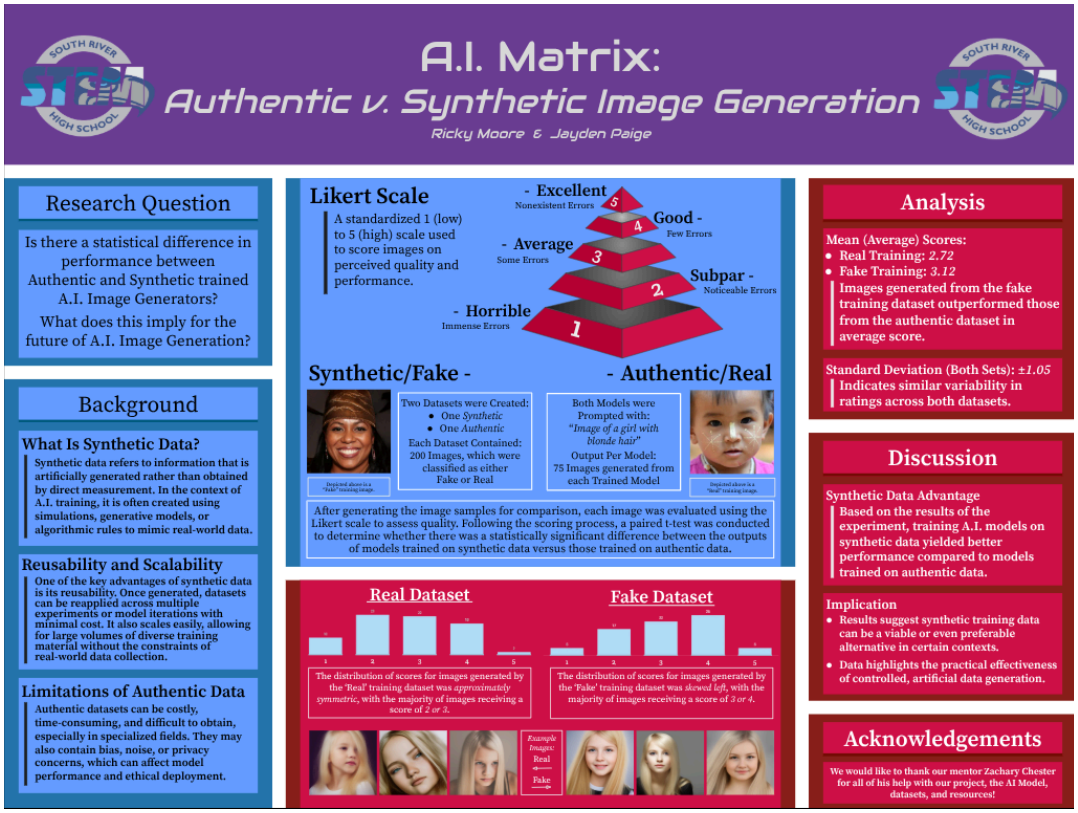
Analysis:

- Paired T-test
- Scoring each of the created images on a Likert scale of 1-5
- Taking the scores and using statology.org for comparison
- Finding out if there is a significant difference or not in the data

Visualizations



Poster


A.I. Matrix:
Authentic v. Synthetic Image Generation
Ricky Moore & Jayden Paige
Research Question
Is there a statistical difference in performance between Authentic and Synthetic trained A.I. Image Generators?
What does this imply for the future of A.I. Image Generation?
Background
What Is Synthetic Data?
Synthetic data refers to information that is artificially generated rather than obtained by direct measurement. In the context of A.I. training, it is often created using simulations, generative models, or algorithmic rules to mimic real-world data.
Reusability and Scalability
One of the key advantages of synthetic data is its reusability. Once generated, datasets can be reapplied across multiple experiments or model iterations with minimal cost. It also scales easily, allowing for large volumes of diverse training material without the constraints of real-world data collection.
Limitations of Authentic Data
Authentic datasets can be costly, time-consuming, and difficult to obtain, especially in specialized fields. They may also contain bias, noise, or privacy concerns, which can affect model performance and ethical deployment.
Likert Scale
A standardized 1 (low) to 5 (high) scale used to score images on perceived quality and performance.
5 - Excellent
Nonexistent Errors
4 - Good -
Few Errors
3 - Average
Some Errors
2 - Subpar -
Noticeable Errors
1 - Horrible
Immense Errors
Synthetic/Fake -
- Authentic/Real
Two Datasets were Created:
One Synthetic
One Authentic
Each Dataset Contained:
200 Images, which were classified as either Fake or Real
Both Models were Prompted with:
"Image of a girl with blonde hair"
Output Per Model:
75 Images generated from each Trained Model
After generating the image samples for comparison, each image was evaluated using the Likert scale to assess quality. Following the scoring process, a paired t-test was conducted to determine whether there was a statistically significant difference between the outputs of models trained on synthetic data versus those trained on authentic data.
Real Dataset
Fake Dataset
The distribution of scores for images generated by the 'Real' training dataset was approximately symmetric, with the majority of images receiving a score of 2 or 3.
The distribution of scores for images generated by the 'Fake' training dataset was skewed left, with the majority of images receiving a score of 3 or 4.
Example Images:
Real
Fake
Analysis
Mean (Average) Scores:
Real Training: 2.72
Fake Training: 3.12
Images generated from the fake training dataset outperformed those from the authentic dataset in average score.
Standard Deviation (Both Sets): ±1.05
Indicates similar variability in ratings across both datasets.
Discussion
Synthetic Data Advantage
Based on the results of the experiment, training A.I. models on synthetic data yielded better performance compared to models trained on authentic data.
Implication
Results suggest synthetic training data can be a viable or even preferable alternative in certain contexts.
Data highlights the practical effectiveness of controlled, artificial data generation.
Acknowledgements
We would like to thank our mentor Zachary Chester for all of his help with our project, the AI Model, datasets, and resources!

Analysis

Authentic Analysis

After scoring the 75 images generated using the “authentic dataset”, a mean score of 2.72 and a standard deviation of 1.05 was calculated. The distribution of scores was skewed slightly right, with more values on the lower end. The most common value was 2, with 23 instances.

Synthetic Analysis

After scoring the 75 images generated using the “synthetic dataset”, a mean score of 3.12 and a standard deviation of 1.05 was calculated. The distribution of scores was skewed slightly left, with more values on the higher end. The most common value was 4, with 26 instances.

Discussion

Based on the results of the experiment, A.I. models trained on synthetic data yielded better performance when compared against models trained on authentic data. Results suggest synthetic training data can be a viable or even preferable alternative in certain contexts. Data highlights the practical effectiveness of controlled, artificial data generation.

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